EARL LOUIS

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EDUCATION

University of Connecticut, Storrs, CT
Bachelor of Science in Robotics Engineering (Electronics Track) | Expected December 2026

TECHNICAL SKILLS

Competencies: Control Panel Design, P&ID Development, Sensor Network Integration, Embedded Systems Design, Power Distribution, Process Automation, SCADA/HMI, PLC Systems

Software & Tools: Ignition (8.1 & 8.3 Credentials), Python (multi-threading, signal processing), C, C++, AutoCAD, KiCad, LTSpice, AVR Microchip Studio

Protocols: Modbus TCP, EtherNet/IP, ROS, I2C, SPI, UART, IoT

PROFESSIONAL EXPERIENCE

Barnes Aerospace, Windsor, CT
Control Systems Intern | June 2025 – Present

- Earned Ignition 8.1 and 8.3 Credentials and configured HMI/SCADA displays for real-time operator data visualization.
- Networked ~10 CNC machines and legacy assets into the plant IoT layer by configuring Ignition tags over Modbus TCP and EtherNet/IP, surfacing live machine and power status on operator dashboards, as part of a two-site rollout spanning 40 to 50 machines under a global program to connect roughly 500.
- Designed a 21-point plant safety monitoring network and led its phased deployment to speed emergency response, auto-dispatching alarms to first-aid, spill-response, and management teams, combining wet-switch detection for leaks and eyewash use with wireless sensors for AED, spill-kit, vacuum-furnace, and part-washer access points.
- Engineered the controls and instrumentation for an upgraded waste tank handling used oil and process water, authoring and reviewing the P&ID and control-panel design documentation for a system integrating level and temperature sensing, immersion heating, and pump interlocks.
- Integrated a pH probe and flowmeters into Ignition for environmental continuous-monitoring compliance, designing the control panel that houses the pH transmitter and building a custom export of true 1-minute interval history (replacing a stock chart limited to roughly 5-minute resolution at 300 interpolated points) for direct submission to the regulating agency.

Norwich Public Utilities, Norwich, CT
Electrical Engineering Intern | June 2024 – January 2025

- Built NPU a reusable analytics layer engineers could query directly, replacing manual per-request data pulls, by consolidating five years of load data across the 4.8 kV and 13.8 kV networks into filterable Excel workbooks and overlaying weather data to expose temperature-driven circuit peaks.
- Resolved a residential voltage-drop issue blocking a customer's EV charging by diagnosing line distance, not transformer capacity, as the root cause and delivering two comparable redesign layouts for the senior engineer's construction decision.
- Validated 41 newly acquired pole-top transformers against tolerance specs through turns-ratio testing, documenting results and flagging out-of-tolerance units for follow-up.
- Surfaced two transformer swap pairs that could rebalance system loading with no new equipment spend, by analyzing load data on roughly 85 over- and under-loaded transformers and mapping them geographically.

Outlier AI, Remote
Machine Learning Quality Assurance Tester | May 2024 – January 2025

- Improved model safety by reviewing roughly 300 Meta AI-generated responses for accuracy, relevance, and safety adherence, flagging and downgrading cases where the model overstepped into medical or mental health advice.

Demers Exposition Services, Inc., Uncasville, CT
Stage Technician | December 2022 – June 2024

- Assisted crew leads with load-in and load-out of large-scale arena events including concerts and sporting events, running audio and lighting cable, staging equipment, and operating chain hoists during rigging setup and teardown.

FEATURED PROJECTS

Drone Health Monitoring System · Raspberry Pi 5, I2C/SPI/UART, FFT, 915 MHz Radio
Joint senior design project. Fitted a quadcopter with current, vibration, and temperature sensors. An onboard Raspberry Pi performed FFT-based processing and transmitted data over 915 MHz radio to a ground station running a live dashboard with automated out-of-range alarms.

Two-Wheeled Path Planning Robot · Romi 32U4, Raspberry Pi 3, Python, C++
Navigation robot with a custom wavefront path planner and overhead camera pipeline for real-time localization and orientation detection.

Sound Intensity Visualizer · AVR 128DB48, C (AVR Microchip Studio), LED Dot Matrix, OLED Display
Audio visualizer displaying live sound waveforms on an LED dot matrix with dB readout on a secondary OLED screen.